# Power Majorants in the Zygmund Whitney Problem: Hölder Cases and Collapse to One Endpoint

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## Status and source question

**Status:** candidate substantial structural partial result, likely valid. Shvartsman [1] defines $C^k\Lambda^m_\omega(\mathbb{R}^n)$ by bounded derivatives through order $k$ and the estimate

$$\|\Delta^m_h D^\alpha f\|_\infty \le C\omega(|h|), \qquad |\alpha| = k.$$

On source PDF page 3, Problem 1.2 asks whether these spaces possess the finiteness property when $m > 2$, or when $m = 2$ and $k > 0$.

An impressive breakthrough in the solution of the Whitney problem for $C^{k,\omega}$-spaces has recently been made by Fefferman [8]-[14]. In particular, one of his remarkable results states that the space $C^{k,\omega}(\mathbf{R}^n)$ possesses the finiteness property *for all* $k, n > 1$, see [8, 10]. (An upper bound for the finiteness number $N(k,n)$ is $N(k,n) \le 2^{\dim \mathcal{P}_k}$, see Bierstone, Milman [3], and Shvartsman [26]. Here $\mathcal{P}_k$ stands for the space of polynomials of degree at most $k$ defined on $\mathbf{R}^n$. Recall that $\dim \mathcal{P}_k = \binom{n+k}{k}$.)

Thus for $m > 2$ or $m = 2$ and $k > 0$ the following problem is open.

**Problem 1.2** *Whether the space $C^k\Lambda^m_\omega(\mathbf{R}^n)$ possesses the finiteness property?*

In this paper we develop an approach to Problem 1.1 which allows us to reformulate this problem as a purely geometric question about the existence of Lipschitz selections of set-valued mappings defined on metric spaces with certain hyperbolic structure.

**2. Extensions of Zygmund functions and Lipschitz selections**

The result below treats all power majorants $\omega(t) = t^s$ with $0 < s < m$. It settles every noninteger power and shows that every integer power is not a new higher-order problem: it is exactly the classical $m = 2$, $\omega(t) = t$ endpoint with a shifted derivative index.

## Proof intuition

For smoothness $s$ strictly below the chosen difference order $m$, the quantity $\sup_h |h|^{-s}\|\Delta^m_h g\|_\infty$ is precisely the usual finite-difference norm of the Hölder–Zygmund/Besov space $B^s_{\infty,\infty}$. Taking $k$ bounded classical derivatives simply shifts Besov smoothness from $s$ to $k+s$. If $s$ is nonintegral this is an ordinary Hölder space, where the finiteness theorem is known. If $s = r$ is integral, the same space can instead be written using $k+r-1$ derivatives followed by one Zygmund derivative; this is $C^{k+r-1}\Lambda_t^2$. The unresolved part is therefore an endpoint selection problem, not a proliferation of distinct higher-difference problems.

## Structural partial theorem

We use inhomogeneous bounded-function norms throughout.

**Theorem 1** (Power-scale classification). *Let $n \geq 1$, $k \geq 0$, $m \geq 2$, and $0 < s < m$. Put $\omega_s(t) = t^s$. Then*

$$C^k \Lambda^m_{\omega_s}(\mathbb{R}^n) = B^{k+s}_{\infty,\infty}(\mathbb{R}^n) \tag{1}$$

*with equivalent norms, with constants depending only on $n, k, m, s$.*

*If $s = a + \theta$ with $a \in \{0, \dots, m-1\}$ and $0 < \theta < 1$, then*

$$C^k \Lambda^m_{t^s}(\mathbb{R}^n) = C^{k+a,\theta}(\mathbb{R}^n), \tag{2}$$

*again with equivalent norms. Consequently this space has the finiteness property.*

*If $s = r \in \{1, \dots, m-1\}$ is integral, then*

$$C^k \Lambda^m_{t^r}(\mathbb{R}^n) = C^{k+r-1}\Lambda^2_t(\mathbb{R}^n) \tag{3}$$

*with equivalent norms. Hence every integer power-majorant instance of Problem 1.2 is equivalent to the classical Zygmund endpoint family.*

*Proof.* The standard finite-difference characterization of inhomogeneous Besov spaces [2, finite-difference and lifting theorems] states that, whenever $0 < s < m$,

$$\|g\|_{B^s_{\infty,\infty}} \asymp \|g\|_\infty + \sup_{0<|h|\leq 1} \frac{\|\Delta^m_h g\|_\infty}{|h|^s}. \tag{4}$$

The restriction $|h| \leq 1$ is immaterial for the source norm, since $\|\Delta^m_h g\|_\infty \leq 2^m \|g\|_\infty$ for every $h$.

The same theory gives the derivative-lifting equivalence

$$\|f\|_{B^{k+s}_{\infty,\infty}} \asymp \sum_{|\beta|\leq k} \|D^\beta f\|_\infty + \sum_{|\alpha|=k} \|D^\alpha f\|_{B^s_{\infty,\infty}} \tag{5}$$

on $C^k_b(\mathbb{R}^n)$. Substitution of (4) into (5) is exactly the norm in the definition of $C^k\Lambda^m_{t^s}$, proving (1).

For nonintegral $s = a + \theta$, the classical identification $B^{k+a+\theta}_{\infty,\infty} = C^{k+a,\theta}$ proves (2). The finiteness property for bounded $C^{q,\theta}$ spaces is part of the Whitney finiteness theory quoted in the source introduction and developed in Fefferman's work [3, 4]. Equivalent norms transfer both the local hypothesis and the global conclusion, so the finiteness property transfers to $C^k\Lambda^m_{t^s}$.

Finally let $s = r$ be integral and put $q = k + r - 1$. Applying (1) first to $(k, m, s) = (k, m, r)$ and then to $(k, m, s) = (q, 2, 1)$ gives

$$C^k \Lambda^m_{t^r} = B^{k+r}_{\infty,\infty} = B^{q+1}_{\infty,\infty} = C^q \Lambda^2_t,$$

which is (3). □

**Corollary 2** (The named $Z_m$ scale). *For every $m \geq 2$,*

$$Z_m(\mathbb{R}^n) := \Lambda^m_{t^{m-1}}(\mathbb{R}^n) = C^{m-2}\Lambda^2_t(\mathbb{R}^n)$$

*with equivalent norms. Thus a positive answer for the source's $m = 2$ endpoint for every derivative index would settle all of its named spaces $Z_m$.*

## Verification, limitations, and upgrade obstruction

The proof uses the finite-difference characterization only in its valid strict range $m > s$. At an integer $s$ it does *not* identify the space with a Lipschitz class; instead it compares two difference descriptions of the same endpoint Besov space, which is why (3) is safe.

This does not settle arbitrary majorants, the borderline case $s = m$, or the integer endpoint $C^q \Lambda_t^2$ for $q > 0$. A direct check of the later sharp Lipschitz-selection theorem found that its range must lie in a Banach space, whereas the source's range is the nonlinear hyperbolic jet space $(\mathcal{P}_L \times \mathcal{K}, d_\omega)$. A Kuratowski embedding does not preserve convexity of the polynomial fibers. A second endpoint route would be a uniformly bounded-cardinality molecular decomposition of finite sums of point evaluations in a $B_{1,1}^{-q-1}$ predual; that decomposition remains unproved here.

Exact-title, arXiv-id, exact-problem-phrase, generalized-Zygmund, and hyperbolic-jet-space searches through 11 August 2026 found no later full resolution. The equivalences used above are classical, so novelty confidence for the structural application is low.

## Human-review recommendation

Verify the inhomogeneous bounded-function convention in (5), and confirm that the selected $C^{q,\theta}$ finiteness formulation has constants independent of the closed trace set. After that, the parameter shift in (3) is formal. For a full upgrade, focus on the nonlinear-bundle finiteness lemma or the endpoint molecular decomposition, not on additional difference-order manipulations.

## References


[1] P. Shvartsman, *The Whitney extension problem for Zygmund spaces and Lipschitz selections in hyperbolic jet-spaces*, arXiv:0905.2602 (2009).

[2] H. Triebel, *Theory of Function Spaces*, Monographs in Mathematics 78, Birkhäuser, 1983.

[3] C. Fefferman, *A sharp form of Whitney's extension theorem*, Ann. of Math. (2) 161 (2005), 509–577.

[4] C. Fefferman, *Whitney's extension problem for $C^m$*, Ann. of Math. (2) 164 (2006), 313–359.